

Homework 7
Fall 1 2024
CSDA 5110

A marketing research firm was engaged by an automobile manufacturer to conduct a pilot study to examine the feasibility of using logistic regression for ascertaining the likelihood that a family will purchase a new car during the next year. A random sample of 33 suburban families was selected. Data on annual family income (X1 in thousands) and the current age of the oldest family automobile (X2, in years) were obtained. A follow-up interview conducted 12 months later was used to determine whether the family actually purchased a new car (Y=1) or did not purchase a new car (Y = 0) during the year.

- i) Fit of logistic regression model.
- ii) Interpret the odd ratios of two estimates of coefficients
- iii) Perform influential diagnostics and identify any influential data points.
- iv) Perform the goodness of fit test.
- v) Obtain confusion matrix, AUC and ROC curve. Interpret the findings.
- vi) Use random forest method to predict Y. Compare its confusion matrix, AUC and ROC curve with that of logistic regression model.

(Y)	(X1)	(X2)
0.0	32.0	3.0
0.0	45.0	2.0
1.0	60.0	2.0
0.0	53.0	1.0
0.0	25.0	4.0
1.0	68.0	1.0
1.0	82.0	2.0
1.0	38.0	5.0
0.0	67.0	2.0
1.0	92.0	2.0
1.0	72.0	3.0
0.0	21.0	5.0
0.0	26.0	3.0
1.0	40.0	4.0
0.0	33.0	3.0
0.0	45.0	1.0
1.0	61.0	2.0
0.0	16.0	3.0
1.0	18.0	4.0
0.0	22.0	6.0
0.0	27.0	3.0
1.0	35.0	3.0
1.0	40.0	3.0
0.0	10.0	4.0
0.0	24.0	3.0
1.0	15.0	4.0
0.0	23.0	3.0
0.0	19.0	5.0
1.0	22.0	2.0
0.0	61.0	2.0
0.0	21.0	3.0
1.0	32.0	5.0
0.0	17.0	1.0